

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Code of Conduct: (192411085)

I N. Mohshitha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Mohshitha
Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5 10marks
Problem Identification	20%	1	1	1	1	1
Debugging	25%	1	1	1	1	1
Optimization	20%	1	1	1	1	1
Analysis	20%	1	1	1	1	1
Code Quality	15%	1	1	1	1	1

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

102/113 Debugging & Optimisation Task

Given

- * IP Address = 192.168.10.65
- * Subnet mask = 255.255.255.192 (126)
- * Gateway = 192.168.10.1

Step 1: Find block size

$$\text{Block Size} = 256 - 192 = 64$$

Subnets:-

- * 192.168.10.0 - 63
- * 192.168.10.64 - 127
- * 192.168.10.128 - 191
- * 192.168.10.192 - 255

Since the IP address 192.168.10.65 falls in the range 64 - 127, it belongs to the Subnet 192.168.10.64/26

Step 2:- Network details

- * Network Address = 192.168.10.64
- * Broadcast Address = 192.168.10.127
- * Valid Host Range = 192.168.10.65 - 192.168.10.126

The configured IP is valid because it lies within the host range

The default gateway 192.168.10.1 belongs to another Subnet, so it cannot communicate with this workstation

Correct gateway: 192.168.10.126 (or any valid router interface in the same subnet).

Usable Hosts: $2^{(32-26)} - 2 = 2^6 - 2 = 64 - 2 = 62 //$

Expanded IPv6 Address

Device A

2001 : 0db8 : 0000 : 0000 : 0000 : f000 : 0042 : 8329

Device B

2001 : 0db8 : 0000 : 0000 : 0001 : 0000 : 0000 : 0001

Since the prefix length is /64, only the first 64 bits represent the network.

Network prefix:

2001 : db8 : 0 : 0 :: /64

Both devices share the same network prefix, so they belong to the same subnet. Communication failure is likely due to interface configuration (or) routing rather.

IPv6 addressing.

Corrat Example

Device A → 2001:db8:0:0::10/64

Device B → 2001:db8:0:0::20/64

Available host address:

$$2^{(128-64)} = 2^{64}$$

= 18,446,744,073,709,551,616 addresses.

3)

Network = 192.168.1.0/24

126

• Network = 192.168.1.0

• Broadcast = 192.168.1.63

• Host range = 192.168.1.1 - 192.168.1.62

Usable Hosts:

$$2^{(32-26)} - 2 = 62$$

/27

• Network = 192.168.1.64

• Broadcast = 192.168.1.95

• Host Range = 192.168.1.65 - 192.168.1.94

121 Usable hosts: $2^{(32-27)} - 2 = 30$

• Network = 192.168.1.96

• Broadcast = 192.168.1.111

• Host Range = 192.168.1.97 - 192.168.1.110

Usable Hosts:-

$$2^{(32-28)} - 2 = 14$$

A department with fewer users should receive a smaller subnet, while departments with more users should receive larger subnets. Using VLSM reduces address wastage and improves utilization.

4) Expand the IPv6 address as in Q2.

Both devices have the network prefix:

2001:db8:0:0::/64

Therefore, they belong to the same subnet.

Suggested correlated addresses:

* Device A = 2001:db8:0:0::1064

* Device B = 2001:db8:0:0::2064

Available host addresses:

$2^{64} = 18,446,744,073,709,551,616$ Addresses

Hence, the subnet provides an extremely large

no. of unique IPv6 host addresses.

Destination IP: 192.168.2.100

Compare it with the routing table

- 192.168.1.0/24 $\rightarrow$ no match
- 192.168.2.0/24 $\rightarrow$ match
- default Route $\rightarrow$ used only if no match exists.

Using the longest prefix match, the router selects.

192.168.2.0/24 $\rightarrow$ Next Hop = 10.0.0.3

The destination 192.168.2.100 lies within the range 192.168.2.1 - 192.168.2.254, confirming that it belongs to the correct subnet.

If communication still fails, verify that next-hop router is reachable, routing entries are correct, interfaces are active, and return route exists. If no matching route is found, the router forwards the packet to the default gateway (10.0.0.1).